

Noncontraction Transport Through Unequal-Length Merges

Collatz proof project — September 5, 2026

Abstract

An unequal-length merging predecessor inherits full noncontraction when its finite prefix is noncontracting and its final coefficient dominates the original prefix coefficient. We formalize this transport principle and an infinite arithmetic family meeting after one versus twelve steps. A bounded exact search supplies additional examples but leaves a global existence obligation. Neither nondivergence nor Collatz is proved.

Let $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n . Write $j_t(n)$ for the odd count and $C_t(n) = 3^{j_t(n)}/2^t$. Call n noncontracting if $C_t(n) \geq 1$ for every $t \geq 0$.

Theorem 1 (Dominating merge). *Suppose n is noncontracting, $T^s(x) = T^t(n)$, $C_i(x) \geq 1$ for $0 \leq i \leq s$, and $C_s(x) \geq C_t(n)$. Then x is noncontracting.*

Proof. For $v \geq 0$, the coefficient cocycle and the shared endpoint give

$$C_{s+v}(x) = C_s(x)C_v(T^s(x)) = \frac{C_s(x)}{C_t(n)}C_{t+v}(n) \geq 1.$$

Earlier times are covered by the prefix hypothesis. The Lean proof uses natural-number cross multiplication and cancellation of a positive power of three. It does not assume the shifted seed $T^t(n)$ is noncontracting; that property does not follow just by shifting the original hypothesis. $\square$

Theorem 2 (An infinite merging family). *For every integer $Q \geq 0$, put*

$$N = 111 + 4374Q, \quad M = 103 + 4096Q.$$

Then $0 < M < N$, and

$$T(N) = T^{12}(M) = 167 + 6561Q, \quad j_1(N) = 1, \quad j_{12}(M) = 8.$$

Every prefix of the twelve-step path from M is noncontracting. Consequently, if N is noncontracting, so is the smaller seed M .

Proof. Lean's kernel verifies $T^{12}(103) = 167$, $j_{12}(103) = 8$ and all thirteen prefix coefficient inequalities. Binary-cylinder transport preserves these parities for $103 + 2^{12}Q$ and adds 3^8Q to its endpoint. The seed N is odd, giving the same endpoint after one step. The coefficient comparison is

$$\frac{C_{12}(M)}{C_1(N)} = \frac{3^8/2^{12}}{3/2} = \frac{2187}{2048} > 1.$$

The first theorem applies, and $N - M = 8 + 278Q > 0$. $\square$

Thus no least positive noncontracting seed can belong to this progression. All N are multiples of three, and the meeting occurs after an increasing forward step. The result is not a construction of a noncontracting seed, nor an exclusion of noncontraction for every member without the minimality condition. No mathematical priority claim is made.

Finite search. The exact Python search takes the 7,495 seeds $0 < n < 2^{18}$ whose first 18 prefix coefficients are at least one. Equal-time/equal-count replacement already covers 1,391 of them as ordinary smaller merges. Searching both forward and replacement lengths from zero through 18 covers another 3,768. All 3,768 additions have certificates satisfying the first theorem; 937 of the selected certificates use a positive forward time. Each is independently replayed, and no original path descends before its meeting. The 2,336 unresolved seeds begin 27, 327, 447, 495, 639, 667, 703, 763. These are finite seed counts, not coverage of all their cylinder lifts. Failure within these bounds says nothing about longer replacements.

Verification and remaining gap. The three public declarations in `Collatz.DominatingMerge` are included in the axiom audit. The reproducible search is `analysis/variable_length_surgery.py`; its tests compare against every smaller seed at depth eight and replay five members of the family. The census is not kernel certified. A proof must still force a suitable smaller predecessor for every hypothetical least noncontracting seed; the present certificates cover only particular cases. Nontrivial cycles also remain outside the conclusion.